Team D13 – The GLayout Visionaries

Team Members

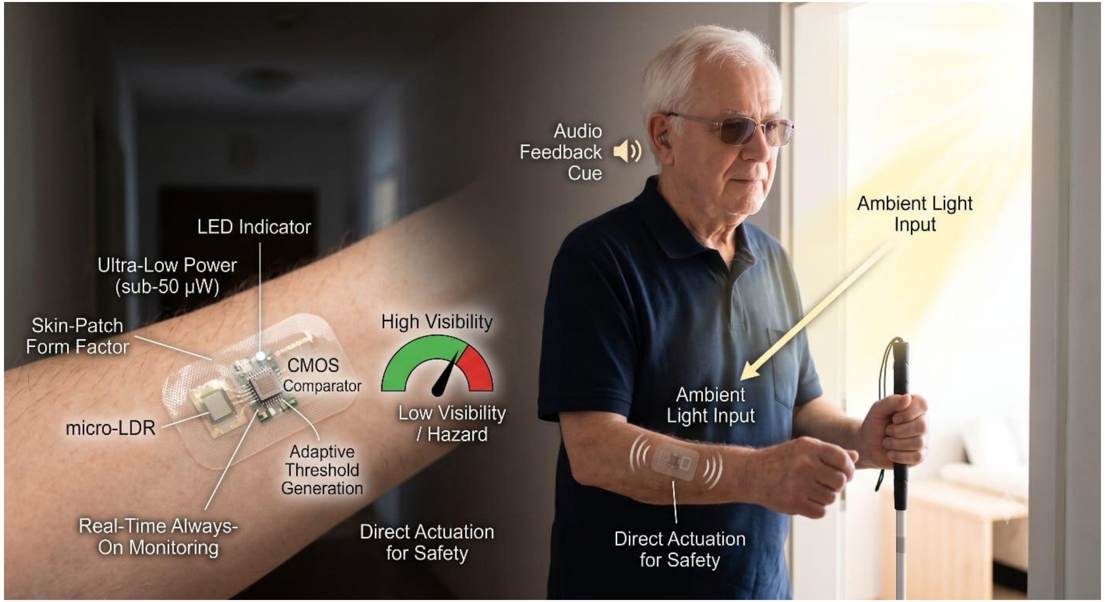
Avinash Pawar – Team Lead | IIIT Surat, PhD Scholar

Bidyut – Member | ALIAH University, BTech Student

Contact email: ai.avinash0007@gmail.com | GitHub: github.com/sscs-ose/sscs-chipathon-2026/issues/124

Project information

Goal: Design a fully integrated CMOS 180 nm analog comparator operating at 1.8 V for wearable skin-patch deployment. The circuit provides real-time binary ambient light classification (HIGH/LOW visibility) with sub-50 µW power consumption and adaptive threshold generation via a resistive divider.



Design – High Level Proposal

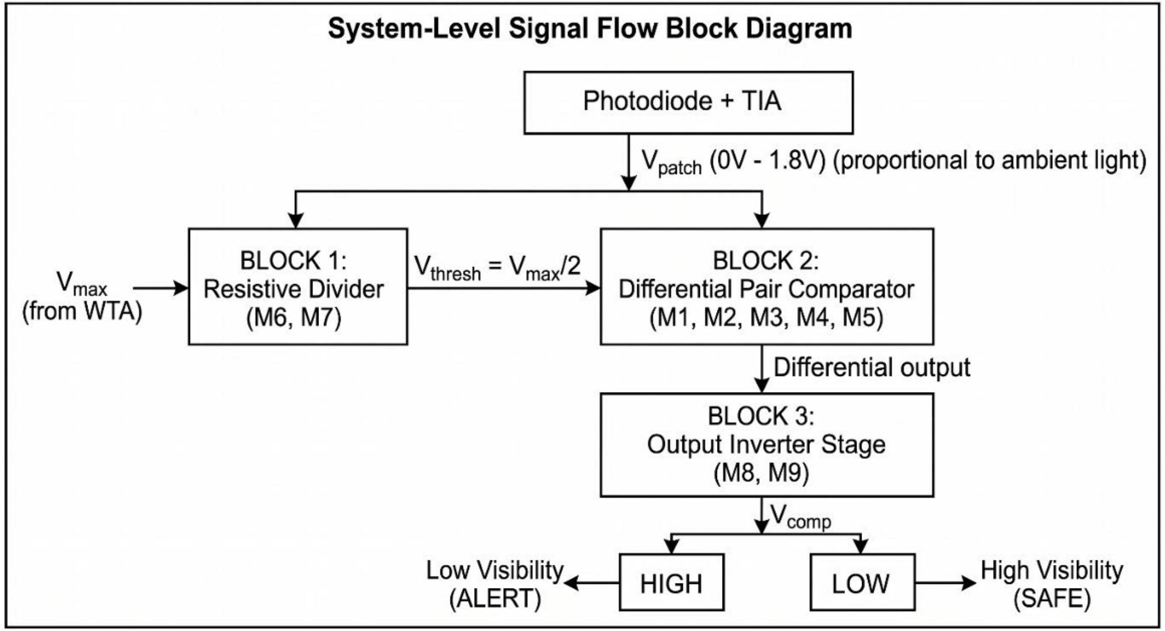
Three-block CMOS comparator chain (9 MOSFETs):

Resistive Divider (adaptive V_thresh = V_max/2) → Differential Pair (M1–M4 + M5 tail) →

CMOS Inverter (M6–M9, rail-to-rail output).

Technology: GF 180 nm PDK (gf180mcu),

open-source EDA via GLayout Framework & NGSPICE simulation.

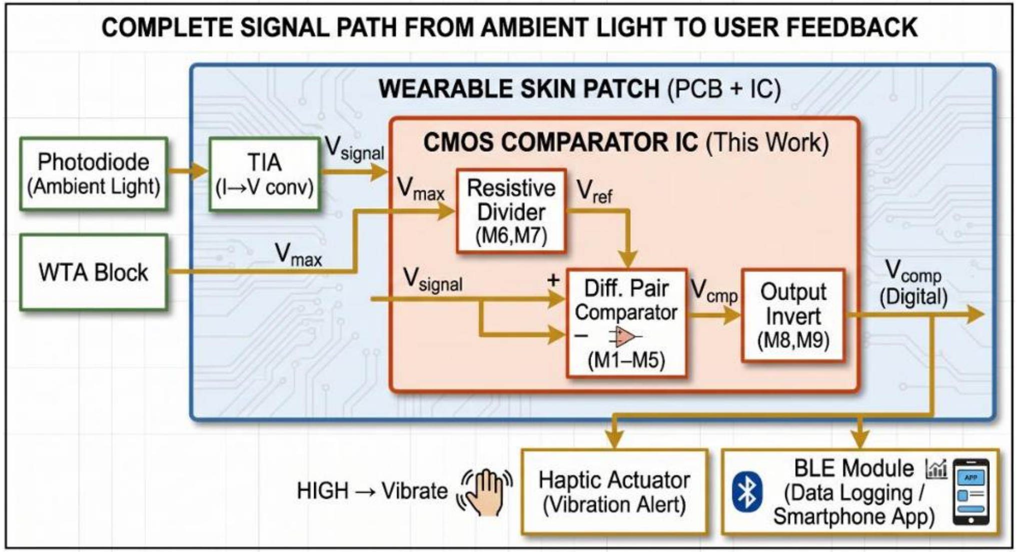


Application

Application:

Wearable assistive device for visually impaired patients — skin-patch light sensor providing haptic/audio alerts for ambient light transitions

(dark corridor, office, outdoor) targeting One year + days battery life on CR2032 at <50 μW total power.



Team background

- Academic Experience:

Avinash Pawar – **PhD Scholar at IIIT Surat;**

- Visiting Researcher, IIT Roorkee (NPTEL Pre-Doc Fellowship 2026);
- IEEE SSCS Member; 37+ GitHub repos;
- Research focus: near-sensor analog processing, AI/LLM-driven circuit optimization (GLayout Framework, SkyWater sky130 / GF180 PDK).

Bidyut – BTech Student,

- ALIAH University;
- Analog/digital circuit fundamentals;
- open-source EDA contributor.

References

1. WHO World Report on Vision, 2019.
2. Stanchieri et al., 1.8V TIA for CMOS optoelectronics, Electronics 2022.
3. Allen & Holberg, CMOS Analog Circuit Design, 3rd ed.
4. Stanchieri et al., 180nm CMOS sensing system, Electronics 2022. Full list: github.com/sscs-ose/sscs-chipathon-2026/issues/124

Questions, Suggestions, Doubts?